RNN

LSTM

↳ NLP - stemming
lemm
NER
} → Current }

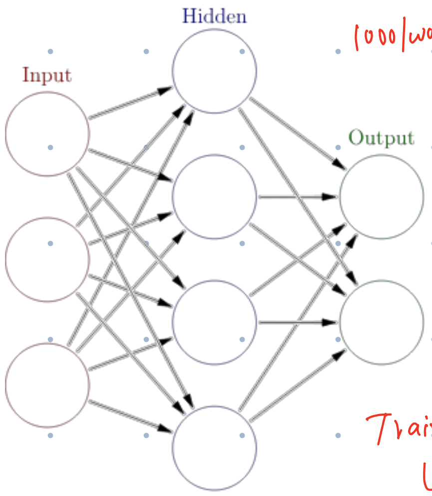
Important Libraries

Attention Mechanism — Parrallely
Transformers

Varying length

ANN → Sequential Data — Text

1000 words



Trained
↳ freezed

→ Context

→ Varying length

→ Parameter sharing

RNN → Recurrent Neural Network

I love
DS, DS
is awesome

o/p = Ci + Po

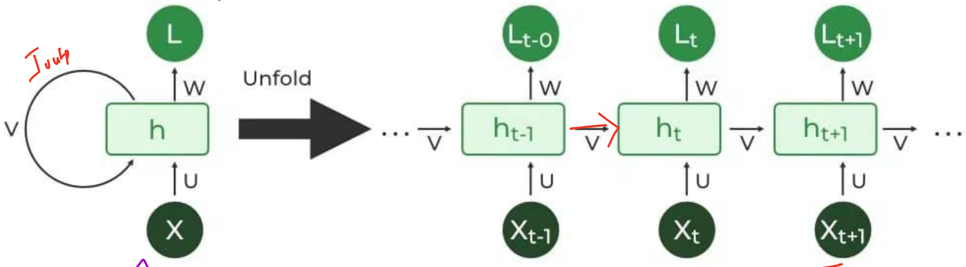
Unfolded View

→ long Sequence

→ Hidden state
↳ dependency

Hidden state

Iout



Sequentially

I/p

Iove

Past

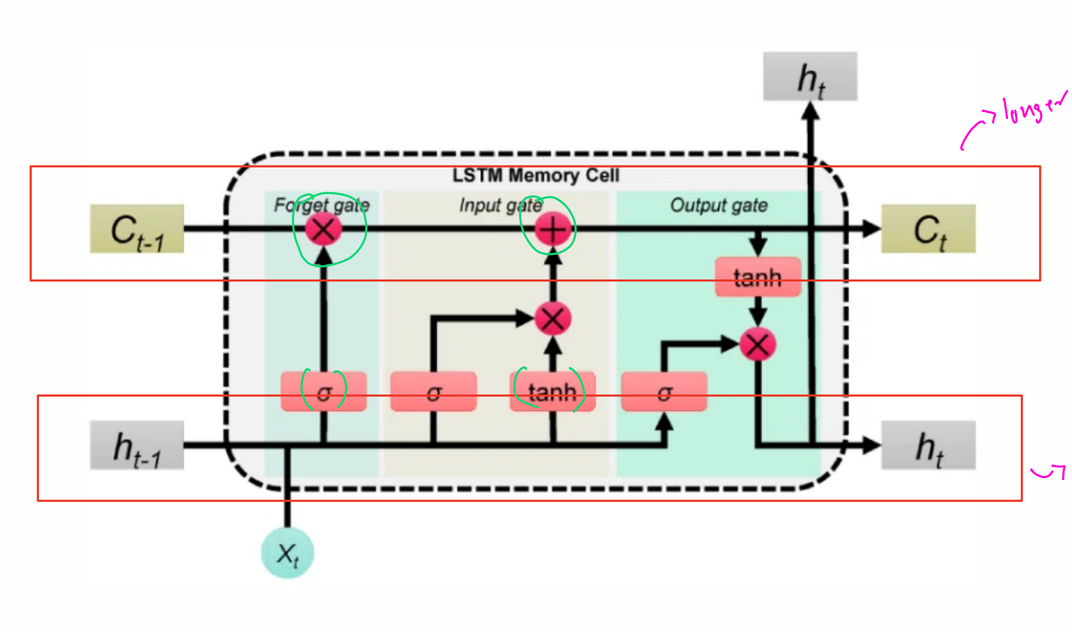
Current
" love "

future

(50)
$0.99 = 0.0$—
zero
$1.01^{50} =$ —

→ long Range Dependency

LSTM → Long Short Term Memory — 15-16 yrs



longer

Shorter

NLP

- human language

- I love this movie → humans

73 20 108 111 118 → See- compute

NLP Pipeline

→ Raw Text → Cleaning → Tokenization → Stop word Removal

→ Stemming / Lemo → Vectorization → ML / DL

↓

Prediction

(1) (I love Data Sci and it's awesomeeee)

(2) awesomeeee

↓

awesome

(3) Tokenization

↳ small piece

(I

love

Data)

(4) Stop word

an

are the

a or

(6) lemmatization

↳ dictionary grammar context

(5) Stemming

↓

cut the word

↳ simple

Porter Stemmer

better

↳ good

running - run

Studies → study

Playing - play

Played - play

Studies → studi → AUC

(8) NER

↳ Named Entity Recognition

(7) Pos (Part of Speech)

The Cat Runs fast

↓ ↳ ↳ ↳ Adverb

Determiner Noun Verb

Person ← Rahul Works in Prodapt Chennai

↓ ↘

Organi location

{ Medical Records Banking

↳ PII

9) Dependendency

↳ Relationship word

(BoW) Bag of words

subject

is object

The Boy Kicked the Ball

Main (Root verb)

Vocabolary

(I Love AI NLP)

Sentence

I Love NLP AI

Vector

Word order is ignored → [1, 1, 1, 1]

10) Vectorization

→ Numbers —

Text

(I) NLTK

(ii) Spacy → Industry

(iii) Textblob → Sentiment

(iv) Gensim → Word2vec

Hugging face → Model

Transformers

Sentence Transformer → Sentence Embedding

Classical NLP

Text

↓

Cleaning

↓

Tokenization

↓

Stop word

↓

Stem

↓

Lem

↓

Modern LLMs

Raw Text

↓

Tokenizer

↓

Token IDS

↓

Embedding

↓

Transformer

↓

Tokens

feature (BRN, pos, ...

↓

ML | DL

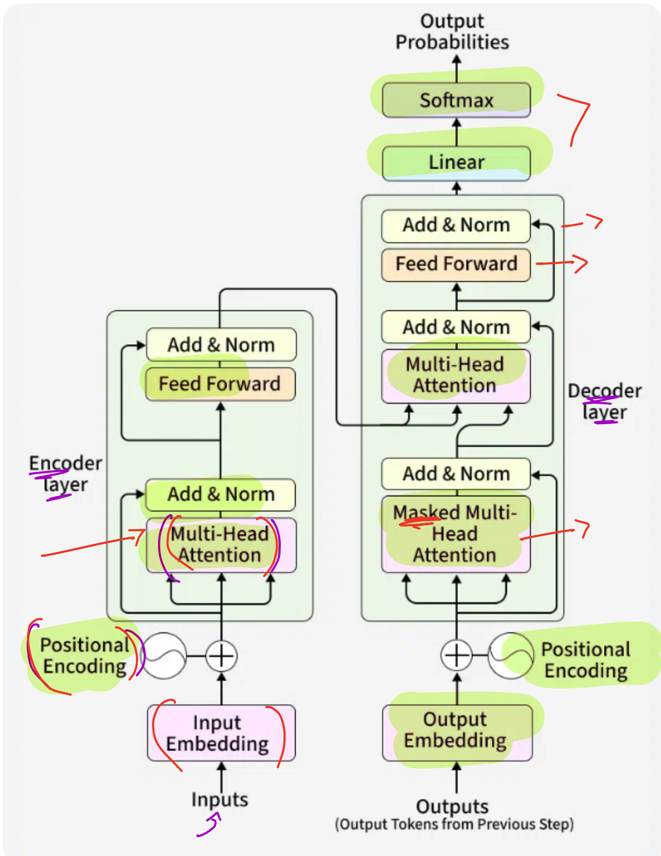
O/p ...

↓

Text

" Attention is All You Need"

↳ 2017 – Vaswani Team



Attention Head

Translation

- ✓ E - F
- ✓ SELF ATTENTION
- ✓ TRansformer
- ✓ Encoder – Decoder

↓ ↓

- ✓ 6 layer 6 layer
- ✓ ED - 512
- ✓ Attention - 8
- ✓ FFN D = 2000+

Encoder Block

Input Embedding

↓

Multi-Head Attention → X 8

↓

Residual connection

↓

layer Normalization

↓

FFN

|

6 X

Decoder Block

Input

↓

Masked Multi-Head Attention

↓

Residual

↓

Cross connections

↓

Residual connection

↓

layer Norm

|

6 X

RC
↓
Layer Noramli
↓
O/P

↓
FFN
↓
layer
↓
O/P

Complete

" The Cat Sat "

→ Tokenization - The
Cat
Sat

→ Embedding → 512
512
512

→ Position Encoding

Embedding + Position
= PoE

→ Self Attention

$Q = XW_Q$
$K = XW_K$
$V = XW_V$

→ Attention

$\mathrm{Softmax}(QK^T / \sqrt{d_k})V$

→ Output
→ Contextual Embedding

→ Residual Connection

Attention Score + Input

→ Layer Norm → FFN – Residual Connec"
↓
Layer Norm
↓
Next Encoder

Data Representation

Natural language

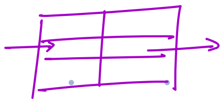
Binary language

Input → processing → output

Binary - 0/1

→ Translate – Compiler

→



Setosa - 0

Versicolor - 1

Virginica - 2

Label Encoder → Text Numerical

→ Chennai → 2 > precedence

" I love Chennai, Chennai is okay okay"

$\Rightarrow$ " I (0) Love (1) Chennai (2) is (3) okay (4) "

OHE → One Hot Encoding

✓ Love → 0 1 0 0 0

Chennai → [0 0 1 0 0] > Solutions

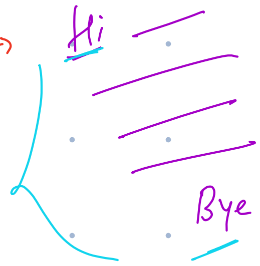
Okay → 0 0 0 0 1

Sparse Vector

– 10,000 words

Huge ← Computation

friend → Hi



Bye

Closer

↳ Zero Semantic Understanding

Embeddings

Text → Numerical

→ Dense Vector

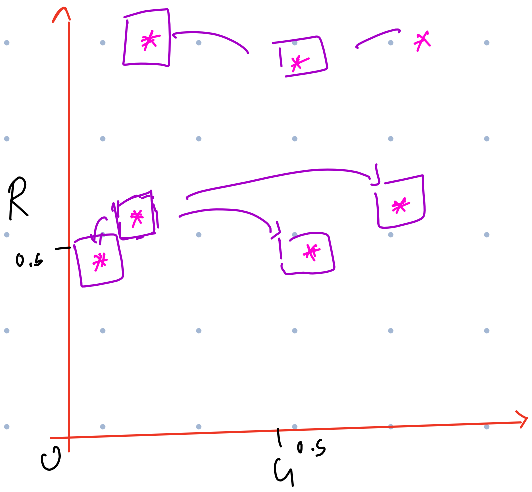
$$\begin{array}{cccc} & R & S & P \\ \text{King} \rightarrow & [0.97, & 0.98, & 0.01] \end{array}$$

Dimensions / features = 3

0 – 1

(0-0.5)(0.5-1)

	Royalty	Gender
King	0.97	0.80
queen	0.98	0.25
Man	0.55	0.75
Women	0.55	0.22
Boy	0.4	0.55
Girl	0.4	0.15
Prince	0.86	0.57



2 Dimension → 1536, 512, 384, 768

3096, 12800+